IRRIGATION WATER MANAGEMENT · NOTE IM04

Micro and Drip Irrigation — Indigenous Devices, Components, Hydraulics, Evaluation and Surge Irrigation

Course topics: IDE7.5 L4, IDE7.5 L5, IDE7.5 L6, IDE-505 T1, IDE-505 T5, IDE-501 T12, CAE1.3 L12, SWCE8.15 L13

Micro-irrigation

Micro-irrigation applies small volumes of water straight to the root zone, wetting only the soil the roots occupy, so surface evaporation and deep percolation are avoided. It includes indigenous devices and modern drip systems, and suits water-scarce areas, widely spaced vegetables, plantation and tree crops and the establishment stage of crops — after which deeper roots can live on stored moisture.

Indigenous and low-cost devices

Device	How it works
Pitcher (pot) method	Earthen pots of 15–20 litres set in a ring basin round the plant, releasing water slowly through cotton wicks in 1 mm holes about 2 cm above the base, or through plastic tubes. Refill the pots as they empty and cover them with straw, dry leaves, coconut husk or plastic sheet to cut evaporation
Modified pitcher	Plastic siphon tubes of 4 mm diameter with a hard-plastic nozzle about 2 cm long and 4 mm across, whose adjustable rectangular opening sets the flow — 2–2.5 L in 24 h per nozzle. One set waters about 4 plants (Fig. 9.28): tubes enter the soil about 5 cm below ground near each plant; the pitcher stands about 25 cm above ground on bricks or straw for head and is filled daily
Sub-soil (root) injector	A hollow tube with a piston and a needle, linked by a pipe to a 15–20 L container, injecting water into the root zone at 15–20 cm depth; the irrigator moves from plant to plant (~30 s each). For porous soils, sloping land, plantations and trees — not clay soils; labour-intensive
Suction irrigation	Uses soil-moisture tension to draw water from buried porous emitters made of clay soil + 30–120% fresh cattle dung (by weight), about 2 cm outside and 3 mm inside diameter. At field capacity tension is 1/10 to 1/3 atmosphere, and suction heads above 1 m can exist in the root zone. Bury the emitters below the root zone, connect them by PVC pipe to a tank, fill to expel air, then lower the tank; a static head H is needed only for the first few minutes. Regulating the negative head matches emission to ET. No external energy (the hydraulic resistance is below the soil suction), airtight joints, and even saline water can be used — salts move away from the roots to the upper wetted soil. For vegetables and fruit trees on undulating land where water is scarce
Mini-sprinklers	Low-pressure under-tree sprinklers for undulating land and long laterals (some with pressure-compensating devices for constant discharge): small volumes over a large area on a low trajectory, set beside the plants so interception is nearly eliminated (or raised above the foliage); less prone to clogging than drippers

Drip (trickle) irrigation

- Slow, localised application to the root zone through pipes and tubes, filters, emitters and ancillaries: water oozes out in drops, so deep percolation and evaporation are minimal. It supplies precise

amounts at frequent intervals — matching daily consumptive use at low soil-moisture tension — and applies fertiliser with the water.

- **History:** developed in Israel (1959) by Simcha Blass (the notes write "Simca"); field research in the early 1960s with sub-surface methods; widespread by 1969 in Israel, the USA, Australia, Mexico and elsewhere, giving higher yields, water savings and good results even with poor-quality water.
- **In India:** widely used in Maharashtra for sugarcane and fruit crops, popular in Tamil Nadu and Karnataka, and spreading to arid and semi-arid regions and to the high-rainfall plantation belt of south-west India, which faces dry-season drought (Nov–May).
- **Key to success:** frequent small doses of water and nitrogen keep a continuous balance of water, air and fertiliser in the root zone — optimum growth, better fruiting and early maturity.
- **Advantages:** saves water; no extensive levelling; higher yields; lower labour cost; fewer weeds; lower salt concentration in the root zone; chemicals and fertilisers through the system; firm, dry inter-row spaces; suits greenhouses; fewer diseases.

Components

Part	Details
Head unit	Pump giving about 2.5 atmospheres; fertiliser tank; filters (with valves and pressure gauge — concentric tanks of pea gravel 0.3–0.5 mm and wire-mesh filters) to stop clogging; a venturi drawing fertiliser into the main by differential pressure
Mains and sub-mains	PVC or HDPE, 4 cm to 11 cm, sized for the design flow; the main conveys all the water, the sub-main adjusts pressure, delivers the required flow to the laterals and controls irrigation time; above ground or buried, with valved outlets for periodic flushing
Laterals (drip lines)	Flexible LDPE or HDPE tubes ≈ 1–1.25 cm (10–20 mm inside), usually one per crop row, up to 40 m long, with about 10% pressure drop allowed along a lateral
Emitters (drippers)	Fixed at regular intervals; small, uniform, constant flow of 2–10 L h^{-1}. Designs: short path, long path, short orifice, vortex, pressure-compensating, self-flushing, perforated double-wall tubing, miniature sprayers. **Point source** (fruit crops), **line source** (closely spaced crops — small fruits, vegetables), **disc source** (micro-sprinklers, sprayers, bubblers). **Micro-tubes** of 0.5–1 mm bore, made uniform by adjusting diameter and length
Control system	Pump, primary filter, secondary on-line filters, flushing units, pressure regulator, pressure gauge, water meter and fertiliser injector

Filter	Use
Screen	The commonest, for solid particles; back-wash arrangement; rising pressure drop signals clogging
Sand (media)	Pressure-type, high-flow; almost the full sand depth is used
Centrifugal	In-line; removes heavier suspended material

System type	Features
Surface	Laterals and emitters on the surface — pressure-compensating, built-in line, adjustable, vortex emitters or 1–4 mm micro-tubing
Sub-surface	Emitters below the surface, including twin-wall (biwall) systems; mainly sugarcane and vegetables
Bubbler	Small stream or fountain; high-volume; not popular in India (under research)
Spray	Spray on the soil surface; not yet popular in India; a possibly cheaper option for close-growing crops

Design and hydraulics

- **Design** the pump and prime mover, head, distribution network, emitters and ancillaries. Pump capacity comes from the crop requirement, spacing, soil moisture, ET climate and differing plant needs; total head = static lift (pumping water level to the highest field point) + lateral operating pressure + losses in main, sub-main, laterals, emitters and fittings.
- **Layout:** emitter positions follow the crop; land slope sets the size and position of mains and sub-mains; soil infiltration, water-holding capacity, texture, structure and bulk density set emitter type, spacing and schedule; climate records set timing and frequency.

$$\text{Blasius: } f = 0.316 \,/\, R_n\text{^}(1/4) \quad \text{Hazen–Williams (main line): } \Delta H = 15.27\ (Q\text{^}1.852 \,/\, D\text{^}4.871)\ L$$

f = friction coefficient; R_n = Reynolds number; ΔH = friction drop (m); Q = discharge (L s^{-1}); L = length (m); D = inside diameter (cm). Because discharge falls along a line with outlets, the actual drop is less than this value, and the energy gradient is an exponential curve, not a straight line.

Missing coefficient. For laterals and sub-mains the notes print ΔH = Q^1.852/D^4.871 with no coefficient or length. The standard multiple-outlet form is **ΔH = 5.35 (Q^1.852 / D^4.871) L** (Q = inlet discharge, L = total length) — i.e. 15.27 × the Christiansen outlet factor of about 0.35. (15.27 itself corresponds to a Hazen–Williams C of about 150, i.e. plastic pipe.)

$$\text{Emitter flow: } q_i = c\,\sqrt{h_i};\quad q_var = (q_max - q_min)/q_max;\quad h_var = (h_max - h_min)/h_max;$$
$$q_var = 1 - (1 - h_var)\text{^}0.5$$

- **Design criteria:** along a lateral, emitter flow variation < 20% (≈ 40% pressure variation); along a sub-main, lateral flow variation < 5% (≈ 10% pressure variation).
- **Pressures:** pump about 2.5 kg cm^{-2} (= 25 m head); 0.5–1.0 kg cm^{-2} lost at the head (filters); about 1 kg cm^{-2} at the lateral inlet; the rest is lost in laterals and nozzles, and water leaves at nearly atmospheric pressure. Nozzle discharge depends on its pressure, opening and internal friction.
- **Spacing:** laterals by crop row and plant spacing; nozzles by discharge and crop — about 1 m for close-growing furrow crops (potato, strawberry); in orchards one nozzle per plant at first, then 2–3 round each tree, 1–1.5 m apart.
- **Laterals** on the surface (easier to check, less clogging, more evaporation at first — the difference shrinks as the canopy grows) or buried 5–10 cm deep. Usually operated daily; interval ≤ 3 days.

Evaluating a drip system

1. Measure the operating pressure near the nozzles with a gauge.

2. Collect nozzle discharge over a long period (L h^{-1}).
3. Find the wetting pattern: run a nozzle 1 h to 24 h or more and read soil moisture (tensiometers or gravimetric) at several depths and distances; plot it.
4. Compute a uniformity coefficient as for sprinklers from the discharges of all nozzles along about six laterals — it depends on lateral length and pressure.
5. Record crop response — growth stage, maturity, yield and quality, pests, diseases and weeds — and study salt movement from soil samples at several depths and points.

Pulse drip irrigation

Water in repeated short ON–OFF cycles produces a wider wetted bulb and less deep percolation; scheduled to the soil, it reduces runoff, percolation and post-irrigation evaporation. Still experimental (mainly with drip), it keeps moisture high for germination. In one device, magnets hold a valve shut while a reservoir charges; the pressure overcomes the magnets, the valve opens, the pressure falls, the magnets close it, and the cycle repeats.

Economics of drip

- Laterals and emitters are the main cost, and their number depends on row and plant spacing — so drip is cheap for orchards and widely spaced vegetables and highly economical in water-scarce regions for fruit, plantation crops, vegetables and sugarcane.
- It does not suit close-growing crops, which need many laterals.

Surge irrigation

- Intermittent irrigation of long furrows (> 25 m up to 200 m) in ON–OFF cycles of 5–10 min during the design duration.
- During ON the water advances; during OFF the wetted soil partly seals and its infiltration rate falls; the next ON advances faster, reaching the tail in 30–50% of the design duration.
- **Advantages:** deep percolation and runoff ≤ 20%; uniform root-zone moisture; irrigation efficiency 85–95%; no criss-cross ridges or feeder-channel divisions, so land lost ≤ 5%.
- **In India:** introduced at TNAU for hydraulic evaluation and crop-compatibility studies. **Limitation:** no marked land or water saving in extremely clayey or extremely sandy soils.

Sources: ICAR eCourse 'Irrigation Engineering' (IIT Kharagpur) — surface, sprinkler and drip irrigation lessons; ICAR eCourse 'Micro Irrigation Systems Design' (AgriMoon). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Typical emitter discharge in drip irrigation is

(a) 0.2–1 L/h
(b) 2–10 L/h
(c) 50–100 L/h
(d) 1 m^3/h

Q2. Drip irrigation was developed in Israel in 1959 by

(a) Simcha Blass
(b) Christiansen
(c) Israelsen
(d) Hazen

Q3. Pump pressure for a drip system is about

(a) 0.25 kg/cm^2
(b) 2.5 kg/cm^2
(c) 10 kg/cm^2
(d) 25 kg/cm^2

Q4. Design limit for emitter flow variation along a lateral is about

(a) 5% (b) 10%

(c) 20% (d) 40%

Q5. For orifice-type emitters q = c√h, so q_var equals

(a) h_var (b) 1 - (1 - h_var)^0.5

(c) h_var² (d) 2 h_var

Q6. Irrigation efficiency of surge irrigation is about

(a) 40–50% (b) 60–70%

(c) 85–95% (d) 100%

Answers

Q1 — (b) 2–10 L/h

Q2 — (a) Simcha Blass

Q3 — (b) 2.5 kg/cm² — *= 25 m head.*

Q4 — (c) 20%

Q5 — (b) 1 - (1 - h_var)^0.5

Q6 — (c) 85–95%